One Difference, Not Two Separate Intervals

AP Statistics · Unit 3 · Topic 3.10 · B: 2026-12-11 · E: 2027-01-22 · TEACHER KEY

CED TETHER

- **Skill 1.D** — Identify an appropriate inference method for confidence intervals.
- **Skill 4.C** — Verify that inference procedures apply in a given situation.
- **Skill 3.D** — Construct a confidence interval, provided conditions for inference are met.
- **EU UNC-4** — An interval of values should be used to estimate parameters, in order to account for uncertainty.
- **LO UNC-4.I** — Identify an appropriate confidence interval procedure for a comparison of population proportions. [Skill 1.D]
- **EK UNC-4.I.1** — The appropriate confidence interval procedure for a two-sample comparison of proportions for one categorical variable is a two-sample z-interval for a difference between population proportions.
- **LO UNC-4.J** — Verify the conditions for calculating confidence intervals for a difference between population proportions. [Skill 4.C]
- **EK UNC-4.J.1** — In order to calculate confidence intervals to estimate a difference between proportions, we must check for independence and that the sampling distribution is approximately normal: a. To check for independence: i. Data should be collected using two independent, random samples or a randomized experiment.
- ii. When sampling without replacement, check that n1 $\leq$ 10%N1 and n2 $\leq$ 10%N2.
- b. To check that the sampling distribution of $\widehat{p}_1$ - $\widehat{p}_2$ is approximately normal (shape): i. For categorical variables, check that n1*$\widehat{p}_1$, n1*(1-$\widehat{p}_1$), n2*$\widehat{p}_2$, and n2*(1-$\widehat{p}_2$) are all greater than or equal to some predetermined value, typically either 5 or 10.
- **LO UNC-4.K** — Calculate an appropriate confidence interval for a comparison of population proportions. [Skill 3.D]
- **EK UNC-4.K.1** — For a comparison of proportions, the interval estimate is ($\widehat{p}_1$ - $\widehat{p}_2$) $\pm$ z* · sqrt($\widehat{p}_1$(1-$\widehat{p}_1$)/n1 + $\widehat{p}_2$(1-$\widehat{p}_2$)/n2). (Note: these formulas can be constructed from the general formula and SE formulas on the AP formula sheet; they need not be memorized.)
- **LO UNC-4.L** — Calculate an interval estimate based on a confidence interval for a difference of proportions. [Skill 3.D]
- **EK UNC-4.L.1** — Confidence intervals for a difference in proportions can be used to calculate interval estimates with specified units.

Essential question: How do the study design and variance structure determine a valid interval for a difference in population proportions?

DOK-3 skill: 4.C · **FRQ pattern:** `two-sample-interval-vs-separate-intervals`

DOK rationale: Part (c) requires adjudicating two plausible interval methods by coordinating the two-sample design, condition evidence, variance structure, confidence-level target, and the inferential consequence of an invalid construction.

Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) **DOK:** 1 → 3 **Minutes:** 45

Teacher does

- Board slide up at the bell. Ask students to choose a method before confirming it.
- Begin the video at minute 5; return at minute 33 to separate variance, SE, and margin of error.
- Collect at the door. Score part (c) only, E/P/I.

Students do

- Choose Elena or Devon and cite design, counts, or zero.
- Complete the follow-along, finish (a)–(c), revise, and turn in the sheet.

Questions to ask

- What single parameter is the town comparison trying to estimate?
- Which four observed counts justify the normal approximation?
- Do standard errors or variances add?

Adult role
Solo teacher. Circulate 5 pairs. Link the formula to design and the rejected method to zero.

[WARN] WATCH FOR

- Pooling the two sample proportions even though the task is to estimate an unrestricted difference.
- Adding separate margins of error instead of adding estimated variance terms under the square root.
- Assuming two individual 95 percent intervals automatically yield a 95 percent interval for their difference.

SCORING PART (c) — E / P / I

Expected elements

- **selects-two-sample-method** (required) — Selects Elena's two-sample z-interval
- **uses-design-and-counts** (required) — Uses independent SRSs and counts 77, 63, 48, and 72
- **explains-variance-combination** (required) — Adds independent variance terms under one square root
- **rejects-separate-interval-coverage** (required) — Rejects automatic 95 percent coverage from two intervals
- **traces-inferential-consequence** (required) — Explains Devon's misleading inclusion of zero

E	Selects Elena, uses design and counts, explains variance and coverage, and diagnoses the false zero conclusion.
P	Selects Elena but incompletely explains the rejected construction or its consequence.
I	Uses separate intervals, pools samples, or ignores design and zero.

[STAR] TODAY'S PROBLEM — annotated

Town Oak has 2,200 households and Pine 1,800. Independent SRSs find 77 of 140 Oak households and 48 of 120 Pine households use curbside composting. Let $p_O - p_P$ mean Oak minus Pine.

Elena: "The two-sample z-interval is $(0.030, 0.270)$."

Devon: "Subtract the farthest endpoints of separate 95% intervals. That gives $(-0.020, 0.320)$, so zero is plausible."

Household samples

Town	Use	Other	n	N
Oak	77	63	140	2200
Pine	48	72	120	1800

FIRST TAKE (5 min)

Which method gives the requested 95% interval for $p_O - p_P$? Cite one design or numerical detail.
Key: Not graded. Equal centers do not make both uncertainty calculations valid.

Rules callout "BUILD ONE INTERVAL FOR THE DIFFERENCE (keep on the page)" is printed on the student sheet.

DOK 1 **(a)** Define p_O and p_P and name the interval procedure for $p_O - p_P$.

Key: p_O and p_P are the proportions of all Oak and Pine households using their town's service. Use a two-sample z-interval for $p_O - p_P$.

DOK 2 **(b)** Check independent samples, both 10 percent conditions, and all four counts. Compute the estimate, SE, margin of error, and 95% interval.

Key: The SRSs are independent; $140 \leq 220$ and $120 \leq 180$; and counts 77, 63, 48, and 72 exceed 10. Thus $\hat{p}_O - \hat{p}_P = 0.55 - 0.40 = 0.15$, $SE = 0.0613829$, and $ME = 1.96(0.0613829) = 0.1203104$. The interval is $(0.0296896, 0.2703104) \approx (0.030, 0.270)$.

DOK 3 **(c)** Choose the warranted method. Use design, counts, variance structure, and zero's location. Explain why separate 95% intervals do not produce the requested coverage and how Devon's result misleads.

Key: Elena is warranted. Independent SRSs and counts 77, 63, 48, and 72 support one two-sample interval. Each sample proportion estimates its own population proportion; independence makes the two estimated variance terms add under one square root. Devon instead combines separate intervals $(0.4676, 0.6324)$ and $(0.3123, 0.4877)$ into $(-0.0201, 0.3201)$. Two individual 95% intervals do not guarantee one 95% interval for their difference. His wider result includes zero and falsely makes no difference look plausible; Elena's requested interval supports a positive Oak-minus-Pine difference of about 3 to 27 points.

EXIT

One thing the video changed about my first take: ______________________________